

Pharmacology Exam: Routes of Drug Administration

Part I: Multiple Choice Questions (MCQs)

Section 1: Factors and Classification

1. Which of the following is NOT a factor determining the selection of a drug administration route?
 - A. Characteristics of the drug.
 - B. Age of the patient.
 - C. The cost of the pharmaceutical company's marketing.
 - D. Condition of the patient (e.g., unconsciousness).
2. Routes of administration are broadly classified into:
 - A. Local and Topical.
 - B. Local and Systemic.
 - C. Enteral and Oral.
 - D. Parenteral and Inhalation.

Section 2: Local Routes

3. Which route is defined as applying a drug to the skin or mucous membrane for local action?
 - A. Topical.
 - B. Sublingual.
 - C. Intravenous.
 - D. Transdermal.
4. Which drug is administered as a tablet but is not absorbed, acting locally for gut sterilization before surgery?
 - A. Paracetamol.
 - B. Neomycin.
 - C. Insulin.
 - D. Nitroglycerin.

5. The intra-arterial route is primarily used for:
- A. Routine vaccinations.
 - B. Diagnostic studies like coronary angiography.
 - C. Treatment of dehydration.
 - D. Managing status epilepticus.

Section 3: Systemic Routes (Enteral)

6. The most common and acceptable route for drug administration is:
- A. Intramuscular.
 - B. Intravenous.
 - C. Oral.
 - D. Inhalation.
7. A major disadvantage of the oral route is that it is:
- A. Painful.
 - B. Not suitable for emergencies due to slow onset.
 - C. Extremely expensive.
 - D. Likely to cause 100% bioavailability.
8. Which drug is kept under the tongue to bypass first-pass metabolism during an acute anginal attack?
- A. Morphine.
 - B. Nitroglycerin.
 - C. Diclofenac.
 - D. Bisacodyl.
9. Which route is used for status epilepticus in children when the drug must be absorbed through the rectal mucosa?
- A. Rectal (Enema).
 - B. Subcutaneous.
 - C. Oral.

- D. Intrathecal.

Section 4: Systemic Routes (Parenteral - Injections & Inhalation)

10. The bioavailability of a drug administered via the intravenous (i.v.) route is:

- A. 50%.
- B. 75%.
- C. 100%.
- D. Variable depending on gastric pH.

11. Which injection route is used specifically for BCG vaccination and drug sensitivity tests?

- A. Subcutaneous.
- B. Intradermal.
- C. Intramuscular.
- D. Intra-arterial.

12. A "depot preparation" for contraception, such as Norplant, is typically inserted into the:

- A. Muscle.
- B. Vein.
- C. Subcutaneous tissue.
- D. Subarachnoid space.

13. Inhalation of drugs like Salbutamol is advantageous because it:

- A. Has a slow onset of action.
- B. Minimizes systemic toxicity.
- C. Is suitable for unconscious patients.
- D. Bypasses the lungs entirely.

14. Which of the following is a disadvantage of the Intramuscular (i.m.) route?

- A. It may cause an abscess or nerve injury.
- B. Absorption is slower than the oral route.

- C. It cannot be used for mild irritants.
- D. It provides 100% bioavailability.

15. What is the correct term for a single, large dose of a drug injected rapidly into a vein?

- A. Infusion.
- B. Bolus.
- C. Depot.
- D. Enema.

Section 5: Transdermal and Special Delivery Systems

16. The transdermal route is beneficial because it:

- A. Is the cheapest route available.
- B. Provides a constant plasma concentration of the drug.
- C. Never causes local irritation.
- D. Has a very short duration of action.

17. Which special delivery system uses minute vesicles made of phospholipids to target drug delivery?

- A. Jet injection.
- B. Liposomes.
- C. Monoclonal antibodies.
- D. Intraoral patches.

Part II: Essay Questions

1. List and explain four factors that influence the choice of a drug administration route.
2. Compare the advantages and disadvantages of the Oral route versus the Intravenous (i.v.) route.

3. Describe the clinical importance of the Sublingual route and provide two examples of drugs administered this way.
 4. Explain the difference between an Evacuant enema and a Retention enema, providing an example for each.
 5. Discuss the precautions that must be taken when administering drugs via the Intravenous (i.v.) route.
 6. Define "Monoclonal Antibodies" and "Liposomes" as special drug-delivery systems and explain their primary benefit.
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Answer Key

Part I: MCQs

1. C | 2. B | 3. A | 4. B | 5. B | 6. C | 7. B | 8. B | 9. A | 10. C | 11. B | 12. C | 13. B | 14. A | 15. B | 16. B | 17. B

Part II: Essay Model Answers (Summary)

1. Factors: Drug characteristics, emergency vs. routine use, site of action (local/systemic), patient condition (unconscious/vomiting), age, and first-pass metabolism.
2. Comparison: Oral is safer, cheaper, and painless but slow and subject to first-pass. IV has 100% bioavailability and quick onset for emergencies, but is invasive, requires asepsis, and cannot be reversed once injected.
3. Sublingual: Quick onset, bypasses first-pass metabolism, and can be terminated by spitting. Examples: Nitroglycerin (angina), Buprenorphine (MI).
4. Enemas: Evacuant enemas (e.g., soap water) are for bowel evacuation. Retention enemas (e.g., methylprednisolone for colitis) are for local or systemic absorption.
5. IV Precautions: Inject slowly, ensure the needle is correctly in the vein, and maintain strict aseptic conditions to avoid phlebitis or tissue necrosis.
6. Special Systems: Liposomes are phospholipid vesicles for targeted delivery. Monoclonal antibodies are cultured immunoglobulins that react with specific antigens. Both are used for targeted delivery, especially for anticancer drugs.

